```
System.out.println("YELLOW - READY");
Thread.sleep(1000);
System.out.println("GREEN - GO");
Thread.sleep(2000);
}
}catch(InterruptedException e){
System.out.println(e);
}
}
}
public class TrafficSignalDemo{
public static void main(String[] args){
TrafficSignal signal=new TrafficSignal();
Thread t=new Thread(signal);
t.start();
}
}
```

Slip 24

1) Write a java program to accept n names of cites from user and display them in descending order.

```
import java.util.*;
public class CityNames{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
System.out.print("Enter number of cities: ");
int n=sc.nextInt();
sc.nextLine();
String[] cities=new String[n];
for(int i=0;i<n;i++)
cities[i]=sc.nextLine();
Arrays.sort(cities,Collections.reverseOrder());
System.out.println("Cities in descending order:");
```

```
for(String city:cities)
System.out.println(city);
}
}
```

2) Create an abstract class Shape with methods area & volume. Derive two classes Sphere (radius), Cylinder (radius, height) from it. Calculate area and volume of both. (Use Method Overriding).

```
abstract class Shape{
abstract void area();
abstract void volume();
}
class Sphere extends Shape{
double radius;
Sphere(double radius){
this.radius=radius;
}
void area(){
System.out.println("Area of Sphere: "+(4*3.142*radius*radius));
}
void volume(){
System.out.println("Volume of Sphere: "+(4.0/3.0*3.142*radius*radius*radius));
}
}
class Cylinder extends Shape{
double radius,height;
Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}
```

```
void area(){
System.out.println("Area of Cylinder: "+(2*3.142*radius*(radius+height)));
}
void volume(){
System.out.println("Volume of Cylinder: "+(3.142*radius*radius*height));
}
}
public class ShapeDemo{
public static void main(String[] args){
Shape s=new Sphere(5);
Shape c=new Cylinder(5,10);
s.area();
s.volume();
c.area();
c.volume();
}
}
```

OR

2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;
public class StudentNames{
public static void main(String[] args){
ArrayList<String> list=new ArrayList<>();
for(String name:args)
list.add(name);
System.out.println("Using Iterator:");
```